Rakan Eyad

AI / Autonomy Engineer
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Summary

Mechatronic engineering graduate working in reinforcement learning and computer vision. Trained RL agents in NVIDIA Isaac Sim for planetary rover navigation, built a real-time computer vision pipeline, and shipped an LLM-based conversation engine to production.

Experience

System Engineer — ALBON Nov 2024 — Jan 2026
- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021
- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021
- Supported server system testing and maintenance in data-centre environments.

Featured Projects

Planetary Rover RL Navigation — Thesis · Isaac Sim
- Developed an Honours thesis on autonomous rover navigation, pairing an A* global planner with a deep-RL local policy trained in NVIDIA Isaac Sim on Mars terrain.
- Implemented a fast A* command term and terrain importer for hybrid planning; tuned SAC training configs and reward/termination logic across 1000s of parallel environments; built a custom evaluation pipeline.

Conjure — Open source · Python
- Built a real-time computer-vision engine classifying 9 hand/face gestures via MediaPipe landmarks, rendering effects at 60 FPS with unit tests and CI.
- Profiled and optimized the render pipeline from 162 ms to 2.8 ms per frame (58x speed-up) through allocation discipline.

Tradie Text — Micro-SaaS · in production
- Built a production micro-SaaS that recovers missed calls for trades businesses using Twilio telephony and a GPT-4o-mini SMS conversation engine.
- Built multi-tenant infrastructure end-to-end: Next.js 15 + TypeScript, Supabase Postgres with row-level security, Redis-backed idempotency/rate limiting, and Stripe usage-based billing.
- Hardened the system solo with 276 CI tests, Playwright e2e coverage, a threat model, and incident runbooks.

Conjure SDK — Open source · Python
- Extracted the engine into a pip-installable SDK with an event-driven API for driving games, overlays, or hardware from gesture events.
- Designed two install tiers (numpy-only and full webcam) with deterministic, webcam-free tests.

Skills

AI & computer vision: RL (SAC/PPO), skrl · PyTorch, Computer vision · MediaPipe, LLM integration (OpenAI), Evaluation pipelines, AI-native / agentic workflows

Software engineering: Python / TS, React · Next.js · Flask, Postgres · Supabase · Redis, Stripe · Twilio integration, pytest · Vitest · Playwright · CI, git · GitHub · profiling

Education

B.E. (Honours), Mechatronic Engineering
University of Sydney · 2022 — 2026
EWAM (Engineering Weighted Average Mark) 76% — Distinction